CASE STUDY 2

Building and Validating a Metal-Organic Framework Synthesis Knowledge Graph with Large Language Models

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CONTENTS

1. Introduction

1.1 Motivation

Metal-organic frameworks (MOFs) are porous crystalline materials assembled from metal nodes and organic linkers. Over 100,000 synthesised structures are recorded in the Cambridge Structural Database, and their applications span CO2 capture, gas storage and separation, catalysis, sensing and drug delivery. The knowledge of how any given MOF is actually made, which metal precursor and which organic linker, in which solvent, by which method, at which temperature and for how long, remains locked in the prose of tens of thousands of publications. Synthesis planning therefore still depends heavily on expert intuition and trial and error.

Large language models (LLMs) and knowledge graphs (KGs) together offer a route from that prose to a structured, queryable resource (Pan et al., 2024). Recent work demonstrates the approach at scale: Bai et al. (2025) built a knowledge graph of 2.53 million nodes for framework materials from more than 100,000 articles, and coupled it to a question-answering system. What that literature does not establish is reliability. How accurate are LLM-extracted synthesis records field by field? How do they compare with the text-mined databases the field already relies on, and with the rule-based systems that produced them? And what does an open-weight model deliver relative to a commercial API, at what cost?

1.2 The gap this work addresses

Three observations define the gap.

First, the existing LLM work on MOF text mining validates against small in-house test sets and uses commercial GPT models almost exclusively (Zheng et al., 2023; Shi et al., 2024; Lin et al., 2025). No published study reports per-field accuracy against both expert annotation and the established text-mined databases.

Second, the authors of DigiMOF, the largest openly available MOF synthesis database, state the problem directly. Writing about routes that papers imply through solvent and temperature rather than naming outright, they observe that such implicit routes "could be easily deduced by a reader but are challenging to extract using rule-based NLP" (Glasby et al., 2023). That is a precise, testable claim about where a language model should help and where it should not.

Third, the cost dimension of the open-weight question is unexamined, despite being the factor that decides whether a pipeline of this kind is reproducible by anyone without an institutional budget.

1.3 Research questions

Main question. How accurately and reliably can large language models extract complete MOF synthesis records, measured against expert annotation and domain-specific baselines?

RQ1. How does per-field extraction accuracy compare between LLMs and a rule-based baseline built from the same domain vocabulary?

RQ2. Which prompting strategy, among zero-shot, few-shot, schema-guided and chain-of-thought, is most reliable for each field of a synthesis record?

RQ3. Where and why do LLM extractions disagree with the DigiMOF and SynMOF reference databases?

RQ4. How do open-weight models compare with commercial APIs on per-field accuracy, cost and latency?

RQ5. Can the resulting knowledge graph answer aggregation queries across hundreds of papers?

1.4 Contributions

1. A reproducible, provenance-complete pipeline from open-access literature to a Neo4j knowledge graph, in which every entity carries a traceable link to the sentence and paper it came from, verified by query rather than asserted.
2. A hand-annotated gold standard of 100 MOF synthesis passages containing 138 triples, drawn by a seeded stratified sample that includes an unflagged control stratum, so the pre-filter's own error rate is measurable rather than assumed.
3. A per-field comparison of ten extractor configurations spanning a rule-based baseline, two commercial models and one open-weight model across four prompting strategies.
4. A pre-registered prediction about where language models should beat rules, recorded before any model was run and confirmed by the measurements.
5. An honest account of what the evaluation cannot show, including a field whose scores reflect annotation granularity rather than extraction quality, and two configurations removed rather than reported from partial data.

1.5 Structure of this report

Chapter 2 reviews the state of the art. Chapter 3 sets out the methodology, including the ontology, corpus construction, annotation protocol and evaluation design. Chapter 4 describes the implementation. Chapter 5 presents the results. Chapter 6 discusses them. Chapter 7 states the limitations and threats to validity. Chapter 8 concludes.

2. State of the Art

This chapter has not been written yet. See `docs/report/README.md` *for what it needs to cover.*

3. Methodology

3.1 Ontology

The extraction target is defined by a MOF-specific ontology (v0.2) held in `configs/ontology.json`, which is the single source of truth for the type system. Nine entity types are defined: MOF, MetalPrecursor, OrganicLinker, Solvent, SynthesisMethod, Condition, Property, Application and Paper. Nine relations connect them:

RELATION	FROM	TO
USES_PRECURSOR	MOF	MetalPrecursor
USES_LINKER	MOF	OrganicLinker
SYNTHESIZED_BY	MOF	SynthesisMethod
IN_SOLVENT	SynthesisMethod	Solvent
AT_CONDITION	SynthesisMethod	Condition
HAS_PROPERTY	MOF	Property
MEASURED_AT	Property	Condition
USED_IN	MOF	Application
MENTIONED_IN	any	Paper

Two design decisions in this table matter later. Solvents and conditions attach to a synthesis method rather than directly to the MOF, so the method acts as a connector; this is what makes the object of those relations the substantive claim. And MENTIONED_IN is never produced by an extractor: it is attached by the pipeline from passage metadata, so that provenance cannot be hallucinated.

The type literals in the extraction interface are checked against the ontology file by automated tests, so the two cannot drift apart silently.

3.2 Corpus

Papers were collected from the Europe PMC open-access subset, which exposes structured JATS full text with labelled sections. This was preferred over publisher scraping and PDF parsing because section labels are given rather than inferred, which the segmentation step depends on.

The query was `(metal-organic framework AND synthesis) AND OPEN_ACCESS:y AND HAS_FT:y AND IN_EPMC:y`. The collected corpus is **399 papers and 20,582,200 characters**, every one carrying a DOI. Licences are 299 CC BY, 56 CC BY-NC-ND, 43 CC BY-NC and 2 unrecognised, the last of which are excluded from redistribution pending verification.

Publisher distribution by DOI prefix is RSC 107, MDPI 78, ACS 75, Nature 40, Wiley 40, Elsevier 11 and a long tail. This matters for RQ3: RSC, ACS and Wiley are the publishers DigiMOF drew from, so an open-access Europe PMC corpus does overlap the reference database's source population.

One paper was returned twice by the search API across cursor pages. Deduplication by PMCID and DOI is applied at collection, because passage identifiers are derived deterministically from the paper identifier and a duplicate paper would otherwise produce colliding passage identifiers in the gold standard.

3.3 Segmentation

The unit of extraction is the synthesis paragraph, not the paper. Papers run 20,000 to 50,000 characters and most of that text contains no recipe, so passing whole papers to an extractor would be both expensive and noisy.

Sections are split into paragraphs with character offsets preserved, short fragments merged into neighbours and long paragraphs split. Each passage receives a stable, deterministic identifier derived from the paper identifier, section name, section occurrence and index, so that the gold standard's references survive a corpus rebuild.

Each passage is scored for synthesis content by a transparent weighted rule set whose weights are module-level constants and sum to 1.0, with saturation caps so that one repeated word cannot carry a passage. The signals are section-title cues, method words, quantity and unit patterns, apparatus terms, MOF solvents and procedure verbs. A learned filter was rejected deliberately: it would need labelled data that this project spends on the gold standard instead, and its decisions could not be recomputed by a reader.

Segmentation produces **22,086 passages**, of which **794 across 224 papers** are flagged as synthesis at the chosen threshold of 0.45, with a mean length of 931 characters.

The threshold is a declared choice, not an accident. Its sensitivity is:

CUTOFF	PASSAGES	PAPERS
0.25	2,369	325
0.35	1,369	267
0.45	**794**	**224**
0.55	485	190
0.65	268	149

3.4 Gold standard

Provenance of the annotations. The gold standard is the instrument against which the language models are measured, so it was produced by the author by hand. Pre-filling it with model output, even from a model not under test, would make the evaluation circular and is the single shortcut this project could not take.

Sampling. The worklist is 100 passages drawn with a fixed seed: 90 from the synthesis-flagged pool and **10 control passages that the pre-filter did not flag**. The control stratum exists so that the pre-filter's own miss rate is measurable; without it, every recall figure reported here would be silently conditional on an unvalidated heuristic. The two strata are interleaved rather than blocked so that annotator fatigue does not fall entirely on one of them.

The sample size was reduced from the 150 to 200 stated in the exposé, on 2026-08-31, because annotator hours are the binding constraint and the submission date is fixed. The cost is precision, reported as wider confidence intervals and reduced support for rare relations. The stratification, the seed and the control stratum were preserved, because those are what make the evaluation reproducible.

Tooling. Annotation used a purpose-built interface in which the relation is chosen first and the subject and object types are then constrained to that relation's legal endpoints, so an ontology-invalid triple cannot be created. Work is autosaved after every action.

Result. 100 records containing **138 triples**. 32 passages contain synthesis records and 68 do not. Per-relation support is:

RELATION	SUPPORT
USES_PRECURSOR	39
IN_SOLVENT	34
AT_CONDITION	34
USES_LINKER	30
SYNTHESIZED_BY	1
HAS_PROPERTY, MEASURED_AT, USED_IN	0

Only the first four relations carry enough support to support a conclusion. The scarcity of the others is itself a finding about where information sits in a paper: properties and applications are rarely stated inside a synthesis paragraph.

3.5 Extractors

All extractors implement one interface and return the same triple shape, so that every strategy is evaluated on identical terms. The interface forbids raising: a failure is recorded in the result rather than aborting a batch.

Rule-based baseline. A dictionary and regular-expression system whose vocabulary is derived from the MOF-adapted ChemDataExtractor parsers used to build DigiMOF and from inspection of the corpus. It was built to work rather than to lose, because a weak baseline would make every reported improvement meaningless. It attempts implicit synthesis routes rather than skipping them, emitting an inferred route at low confidence with a null span so that explicit and inferred routes can be scored separately.

Language models. Four prompting strategies (zero-shot, few-shot, schema-guided, chain-of-thought) are defined as versioned template files rather than inline strings, so the template version forms part of the response cache key. Each template states the allowed entity and relation types with their endpoint constraints and forbids inventing values not present in the passage. Responses are parsed defensively: triples with invalid relation types or endpoint violations are dropped with the reason recorded rather than admitted.

Models. gpt-4o and gpt-4o-mini as the commercial strand, and qwen3.8-27b served on a free tier as the open-weight strand. The exposé named Llama-3; that model is no longer hosted by the provider used here, which is recorded in Chapter 7 as a deviation.

3.6 Evaluation design

Matching is per (passage, relation) group with greedy one-to-one assignment, so no prediction can be credited outside its own passage and no gold triple can be matched twice. Scoring is binary: a pair counts as a true positive or it does not.

Two matching modes are implemented, exact and relaxed, both built on a single shared normaliser that is also used by the graph loader, so that the evaluation and the artefact cannot disagree about whether two names denote the same chemical. The normaliser resolves documented synonyms, drops hydrate notation, unifies unit spelling without converting magnitudes, and deliberately keeps UiO-66 distinct from UiO-67.

One concession is declared. IN_SOLVENT and AT_CONDITION are scored on the object alone. The annotation tool contained a defect in which a text field retained its previous value when the relation changed, so every gold triple of these two types records the MOF's name in the subject position where the synthesis method belongs. Scoring those subjects literally would mark a model wrong for correctly answering "solvothermal". The concession is independently justified by the ontology, in which the subject of these relations is a connector rather than a claim, and it is carried on every evaluation result so that a generated table states it. What it costs is that this study cannot show whether a model attaches a solvent to the correct synthesis when a passage describes several.

4. Implementation

This chapter has not been written yet. See `docs/report/README.md` for what it needs to cover.

5. Results

All figures below were produced by the pipeline in this repository and can be regenerated with `bash scripts/run_experiments.sh` followed by `python -m src.evaluation.run_eval --mode relaxed`. Ten extractor configurations were run over 100 gold passages containing 138 annotated triples, for a total API spend of 3.06 USD. Every configuration reported here covers all 100 passages; two were removed rather than reported from partial data, as described in Section 5.6.

5.1 The pre-registered prediction

Before any language model was run, the following prediction was recorded in `docs/baseline_findings.md`, on the basis that conditions and solvents are matched by local surface patterns that rules already handle well, whereas identifying which material is being made is not:

> *The LLM margin over the rule baseline should be largest on USES_PRECURSOR and USES_LINKER, and smallest on AT_CONDITION and IN_SOLVENT.*

Measured, comparing the best language-model configuration with the rule baseline:

FIELD	RULE BASELINE F1	BEST LLM F1	MARGIN	PREDICTED RANK
USES_PRECURSOR	0.13	0.55	+0.42	largest, confirmed
USES_LINKER	0.26	0.57	+0.31	large, confirmed
IN_SOLVENT	0.23	0.48	+0.25	smaller, confirmed
AT_CONDITION	0.00	0.17	+0.17	smallest, confirmed

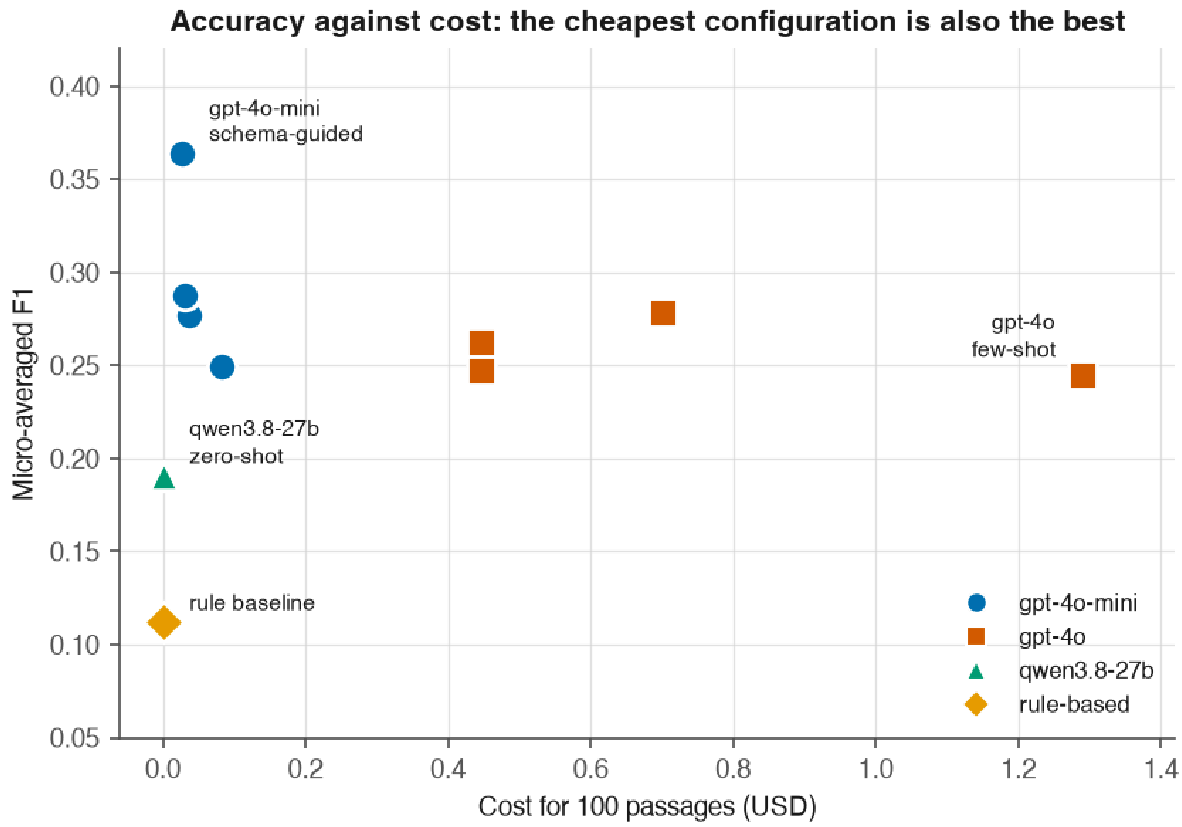
The ordering matches the prediction exactly. Because the prediction was committed to the repository before the models were run, this is a confirmation rather than a pattern identified after the fact.

5.2 Overall performance

Relaxed matching, micro-averaged over all scored relations:

EXTRACTOR	PRECISION	RECALL	F1	COST (USD)	MEAN LATENCY
gpt-4o-mini schema-guided	0.310	0.442	**0.364**	0.028	3.3 s
gpt-4o-mini zero-shot	0.249	0.341	0.287	0.030	3.8 s
gpt-4o chain-of-thought	0.281	0.275	0.278	0.701	5.8 s
gpt-4o-mini chain-of-thought	0.271	0.283	0.277	0.037	5.2 s
gpt-4o schema-guided	0.245	0.283	0.263	0.446	6.5 s
gpt-4o-mini few-shot	0.199	0.333	0.249	0.083	4.5 s
gpt-4o zero-shot	0.256	0.239	0.247	0.447	3.2 s
gpt-4o few-shot	0.226	0.268	0.245	1.289	8.0 s
qwen3.8-27b zero-shot (open weight)	0.136	0.319	0.191	0.000	13.2 s
rule-based baseline	0.098	0.130	0.112	0.000	0.001 s

Every language-model configuration outperforms the rule-based baseline. The weakest language model configuration scores 0.191 against the baseline's 0.112.



Free-tier models are plotted at zero marginal cost. Single run, temperature 0.

Figure 1. Accuracy against cost. The relationship is the finding: the strongest configuration sits at the far left, and the most expensive configuration scores lower than one costing a forty-seventh as much. Free-tier models are plotted at zero marginal cost. Marker shape encodes model family in addition to colour, so the figure survives greyscale reproduction.

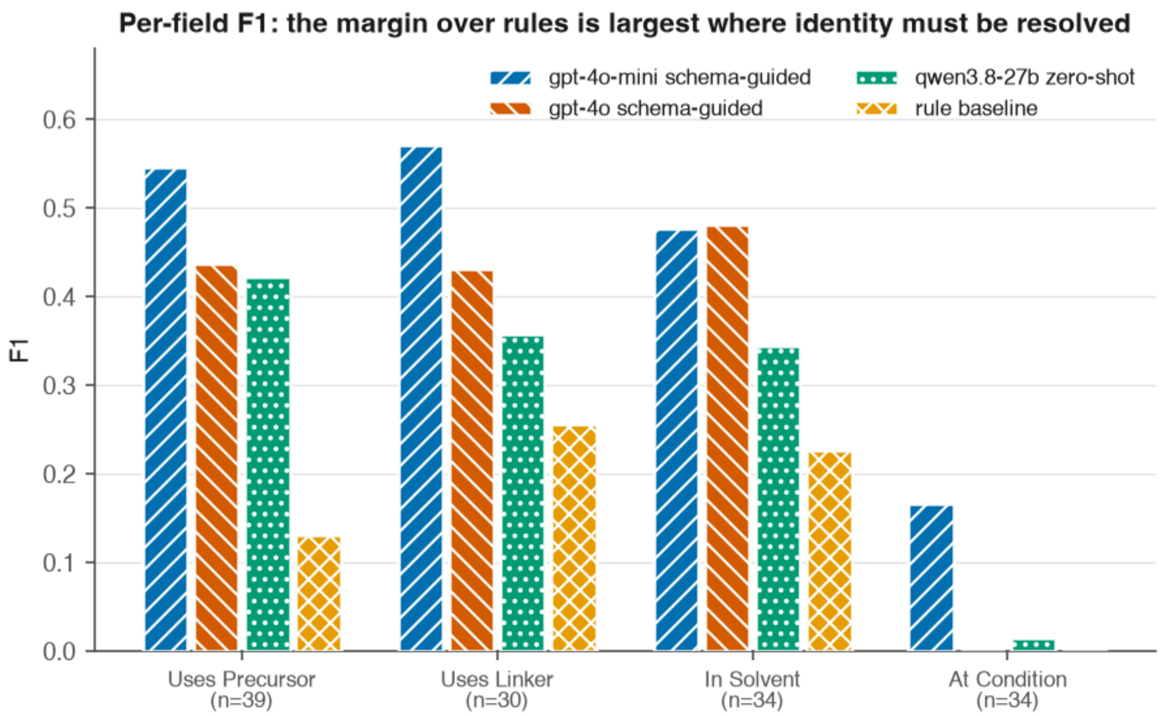
5.3 Per-field results (RQ1)

For the strongest configuration, gpt-4o-mini with schema-guided prompting:

FIELD	P	R	F1	TP	FP	FN	SUPPORT
USES_LINKER	0.615	0.533	0.571	16	10	14	30
USES_PRECURSOR	0.553	0.538	0.545	21	17	18	39
IN_SOLVENT	0.485	0.471	0.478	16	17	18	34
AT_CONDITION	0.140	0.206	0.167	7	43	27	34

Support counts are printed beside every score deliberately. SYNTHESIZED_BY has a support of 1 and HAS_PROPERTY, MEASURED_AT and USED_IN have a support of 0, so no conclusion is drawn about them; where a model emits triples for those relations they appear only as false positives, which is why HAS_PROPERTY shows 21 false positives against zero support.

Excluding AT_CONDITION, whose behaviour is explained in Section 5.5, the best configuration averages approximately 0.53 across the three remaining fields.



n is the number of gold triples for that field. AT_CONDITION scores reflect an annotation granularity mismatch, see Chapter 7.

Figure 2. Per-field F1 for four representative configurations, with the number of gold triples printed on each axis label. Support is shown in the figure rather than only in the caption so that fields backed by different amounts of evidence are not compared silently.

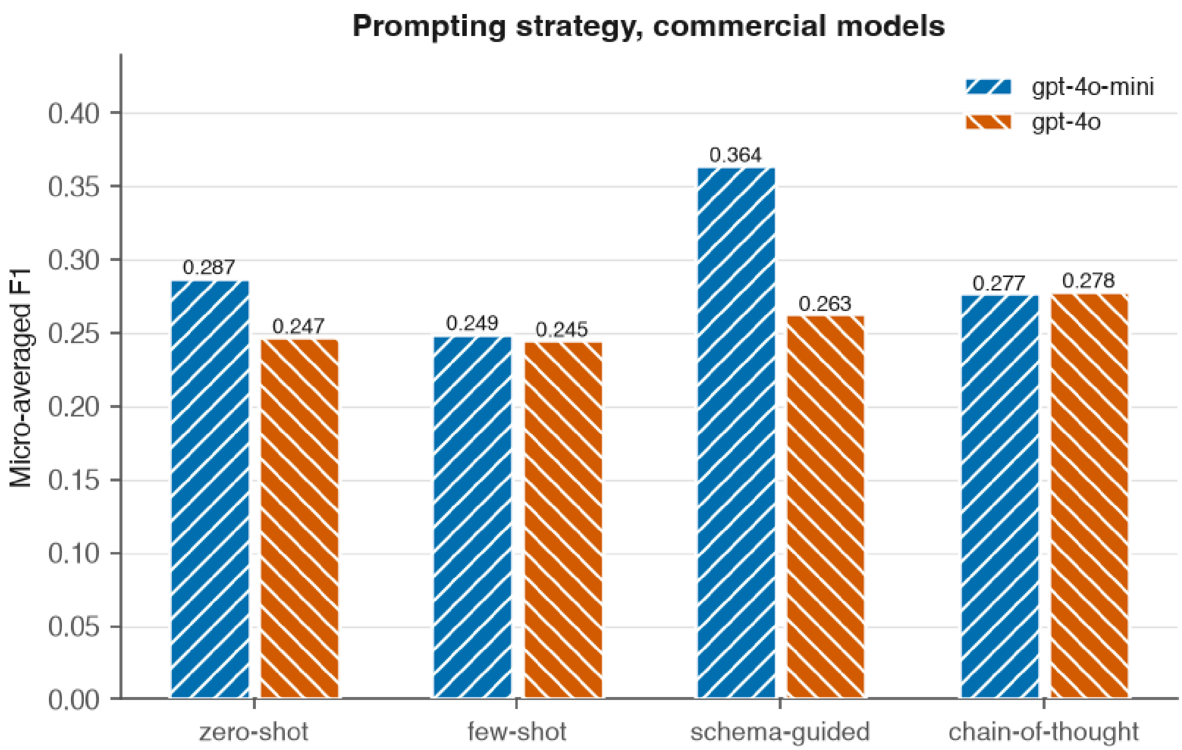
5.4 Prompting strategies (RQ2) and open weight against commercial (RQ4)

Prompting strategies, commercial models, all four strategies complete:

MODEL	ZERO-SHOT	FEW-SHOT	SCHEMA-GUIDED	CHAIN-OF-THOUGHT
gpt-4o-mini	0.287	0.249	**0.364**	0.277
gpt-4o	0.247	0.245	0.263	**0.278**

Schema-guided prompting wins clearly on the stronger configuration, supporting the premise that constraining output to the ontology helps. It does not win universally: on gpt-4o, chain-of-thought leads by 0.015, a margin a 138-triple gold standard cannot resolve. The defensible claim is that schema-guided is the best single choice observed, not that it dominates.

Few-shot is the weakest strategy on both models despite having by far the most expensive prompt, 4,054 template tokens against 652 for schema-guided. The worked examples cost roughly six times more per call and bought nothing measurable here.



The open-weight model is omitted: only one of its four strategies completed within the free-tier token cap.

Figure 3. Prompting strategy by model. Only the two commercial models appear: just one of the open-weight model's four strategies completed within the free-tier token cap, and placing a single point beside two complete sets would invite a comparison the data cannot support.

Open weight against commercial, compared like for like on zero-shot, the one strategy where all three models have complete coverage, so that model and strategy are not confounded:

MODEL	F1	COST (USD)	MEAN LATENCY
gpt-4o-mini (commercial)	0.287	0.030	3.8 s
gpt-4o (commercial)	0.247	0.447	3.2 s
qwen3.8-27b (open weight)	0.191	0.000	13.2 s
rule-based baseline	0.112	0.000	0.001 s

The ordering is commercial, then open weight, then rules. The open-weight model reaches approximately two thirds of the best commercial F1 at zero marginal cost and roughly four times the latency, the latter being a property of the free serving tier rather than of the model.

The cheaper commercial model outperforms the more expensive one. gpt-4o-mini beats gpt-4o on three of four prompting strategies. The strongest configuration overall costs 0.028 USD; the most expensive configuration, gpt-4o few-shot, costs 1.289 USD and scores lower, a 47-fold cost difference in favour of the weaker model. Chapter 6 discusses the likely mechanism.

5.5 Knowledge graph (RQ5)

Loading the rule-based extractions over all 794 synthesis passages produced a graph of **485 distinct nodes and 2,429 relationships across 182 papers**, from 1,864 triples with zero skipped and zero load errors. Entity resolution collapses 3,728 node merge operations into 485 distinct nodes.

Provenance is verified rather than asserted. The `provenance_violations` query, which returns any entity lacking a MENTIONED_IN edge to a source paper, returns zero rows.

Cross-paper aggregation over synthesis methods:

METHOD	PAPERS	MOFS
solvothermal	61	30
hydrothermal	43	12
electrochemical	36	6
microwave-assisted	36	6
stirred at room temperature	35	5
reflux	21	3

DigiMOF reported more hydrothermal (5,677) than solvothermal (3,672) records and described that ordering as surprising, since solvothermal is the more common laboratory route. This corpus shows the opposite ordering. The difference may reflect corpus composition, an open-access Europe PMC set against a CSD-derived one, or a difference in how each system infers an unnamed route. It is recorded here as an open question rather than resolved.

5.6 What is not reported, and why

Three of four open-weight prompting strategies could not be completed. The free serving tier caps this model at 200,000 tokens per day and 8,000 per minute. The per-minute throttle is handled by retrying on the provider's stated delay; the daily cap is a hard stop. Schema-guided completed 26 of 100 passages before exhaustion and few-shot completed 3 of 100.

Both were deleted from the results rather than scored. A configuration answering 26 of 100 passages is still scored against all 138 gold triples and reports an artificially low recall, which would misrepresent a rate-limited model as a weak one. Reporting a strategy as not obtained is honest; reporting it as scoring near zero is not.

RQ4 therefore rests on a single prompting strategy for the open-weight side.

6. Discussion

This chapter has not been written yet. See `docs/report/README.md` *for what it needs to cover.*

7. Limitations and Threats to Validity

This chapter states what the study cannot show. Several of these limitations were discovered during the work and are recorded with the evidence that produced them, so that a reader can judge their severity rather than take the assessment on trust.

7.1 The AT_CONDITION scores measure annotation granularity, not extraction quality

AT_CONDITION scores between 0.00 and 0.17 for every extractor, with 43 false positives for the best configuration. The cause is a granularity mismatch, not extraction failure. The annotator recorded a complete condition string as one value, for example "100 °C for 1 h, then additional 1 h", while the models emit one triple per condition, "100 °C" and "1 h" separately. Neither convention is wrong and the metric punishes the mismatch.

Consequence. No claim about condition-extraction accuracy in this study is trustworthy. The field is reported for completeness and excluded from conclusions. Excluding it, the best configuration averages approximately 0.53 across the three remaining fields.

Remedy for future work. Either fix the annotation convention to one condition per triple, or implement a set-valued comparison for this field that credits a decomposition of a combined gold value.

7.2 Absolute scores are depressed by surface-form mismatch

Manual inspection of gold against predicted triples shows correct extractions scored as failures because of surface form. Examples observed:

GOLD	PREDICTED	STATUS
$ZrOCl_2 \cdot 8H_2O$	`ZrOCl2·8H2O`	matched, the normaliser handles Unicode subscripts
`DMF`	`dimethylformamide (DMF)`	matched only after a fix, see 7.3
`[emim]Br / bmim`	`1-ethyl-3-methylimidazolium bromide`	not matched, same reagent, absent from the synonym table

The measured F1 of 0.364 is therefore a lower bound on semantic extraction quality. A study with a chemical-name resolution service, for example one backed by PubChem or ChemSpider identifiers, would report higher numbers for identical extractions. This limits comparability with published figures obtained under different matching regimes.

7.3 One normalisation change was made after seeing the data

`normalize_chemical` was extended to resolve "full name (ABBREV)" through either half after inspection of failures revealed the pattern. This is disclosed rather than folded in silently. The change raised every extractor by between 0.006 and 0.027, including the rule baseline, and both sets of numbers are preserved in the repository history. A uniform lift across all systems is consistent with a genuine surface-form fix rather than tuning toward a target, but the reader should know the change was made in response to the data.

7.4 The gold standard is small, and smaller than planned

The exposé specified 150 to 200 annotated passages; 100 were annotated, yielding 138 triples. The reduction was a deliberate response to a fixed submission date and the fact that the gold standard must be produced by hand. The consequence is wide confidence intervals on every per-field score and insufficient support for four of the eight relations. Differences of the order of 0.015, such as the schema-guided against chain-of-thought gap on gpt-4o, cannot be resolved at this sample size and are not treated as findings.

Single-annotator annotation is a further limitation: no inter-annotator agreement statistic can be computed, so the gold standard's own consistency is unmeasured.

7.5 A defect in the annotation tool required an evaluation concession

The annotation interface retained a text field's previous value when the relation changed while its label changed underneath. Consequently every IN_SOLVENT and AT_CONDITION gold triple, 68 of 138, records the MOF's name in the subject position where the synthesis method belongs. The tool was fixed, but the recorded gold standard carries the artefact.

The evaluation therefore scores those two relations on the object alone. This is independently defensible, because the ontology routes solvents and conditions through a method that acts as a connector rather than a claim, but it means the study cannot show whether a model attaches a solvent to the correct synthesis when a passage describes several. Correcting the gold subjects with a model was rejected, since generating ground truth with a language model would make the evaluation circular.

7.6 The open-weight comparison rests on one prompting strategy

Free-tier token caps prevented three of four open-weight strategies from completing. RQ4 is therefore answered on zero-shot only. The comparison is like for like, since all three models have complete zero-shot coverage, but it cannot show whether the open-weight model responds to prompting strategy in the same way the commercial models do.

The exposé specified Llama-3 via a local Ollama installation. That plan was abandoned twice: first because 8 GB of RAM made several thousand local inferences impractical within the schedule, and then because the chosen hosted provider had retired every Llama chat model by the time of the run. The substitute, qwen3.8-27b, is genuinely open-weight and independent of the commercial models' lineage, but it is not the model the exposé named.

7.7 Corpus scope and its effect on RQ3

The corpus is restricted to the Europe PMC open-access subset. Much open-access chemistry is not indexed there, so the corpus is not a random sample of MOF literature. The publisher distribution does overlap DigiMOF's sources, which supports the feasibility of the agreement analysis, but coverage bias remains.

The DigiMOF supporting information is keyed by CSD refcode while this corpus is keyed by DOI, and a per-paper join requires a refcode-to-DOI mapping that is normally obtained from a licensed resource. RQ3 is consequently not answered in this report and is carried as future work.

7.8 Single run, no variance estimate

Every configuration was run once, at temperature 0. No variance across repeated runs is reported, so apparent differences between configurations include an unmeasured sampling component even at temperature 0, where provider-side non-determinism can still occur.

8. Conclusion and Outlook

8.1 What was built

An end-to-end, reproducible pipeline that collects open-access MOF literature, segments it into synthesis passages, extracts structured synthesis records with either a rule-based system or a language model under four prompting strategies, evaluates those extractions per field against a hand-annotated gold standard, and loads them into a Neo4j knowledge graph in which every entity is traceable to the sentence that produced it. Provenance completeness is verified by query rather than asserted.

8.2 What was found

Language models beat the rule-based baseline on every configuration tested, and they beat it by the largest margin exactly where a pre-registered prediction said they would: on identifying which material is being made and from what. The margin is +0.42 F1 on metal precursors and +0.31 on organic linkers, falling to +0.25 on solvents and +0.17 on conditions, which are the fields local surface patterns already handle.

The cheaper commercial model outperformed the more expensive one on three of four prompting strategies, at a 47-fold cost difference in favour of the cheaper model. The most likely mechanism, that a larger model produces more elaborate output and therefore more false positives against a fixed gold standard, is consistent with the precision figures but is not established here.

Schema-guided prompting was the best single choice observed, though not universally, and few-shot prompting was the weakest strategy on both commercial models despite the most expensive prompt by a factor of six.

The open-weight model reached approximately two thirds of the best commercial F1 at zero marginal cost. For a pipeline that must be reproducible without an institutional budget, that is a meaningful result rather than a shortfall.

Absolute accuracy is well below the target stated in the exposé. The best configuration reaches 0.364 micro-F1 against a target of 0.80. Chapter 7 sets out how much of that gap is attributable to surface-form matching and annotation granularity rather than extraction quality, and reports that the best configuration reaches approximately 0.53 when the field whose scores measure granularity is excluded.

8.3 What this says about the field

The DigiMOF authors observed that synthesis routes implied rather than named are hard for rule-based extraction. This work provides a quantified version of that observation from the other direction: the rule-based baseline identified a MOF in only 15 percent of synthesis passages, and because the ontology roots five relations at the MOF node, that single failure suppresses most of what a rule-based system could otherwise extract. Of the passages where no MOF was identified, 331 named it elsewhere in the same paper and 251 used a generic designation such as "compound 1". Those are not lexicon gaps that a larger dictionary would close.

8.4 Future work

1. **Complete RQ3.** Obtain a DOI-keyed export of the DigiMOF text-mined table, or accept a MOF-name join with its ambiguity stated, and run the agreement analysis.
2. **Complete the open-weight strategy sweep**, which requires either a paid serving tier or several days of free-tier quota.
3. **Repair the condition evaluation**, either by changing the annotation convention to one condition per triple or by implementing set-valued comparison for that field.
4. **Add chemical identifier resolution** so that surface-form variation stops being scored as extraction error.
5. **Enlarge the gold standard and add a second annotator**, which would both narrow the confidence intervals and allow an inter-annotator agreement statistic.
6. **Investigate the cheap-model result** with a manual error review, since it is the most practically consequential finding here and currently rests on an unverified mechanism.

9. References

This chapter has not been written yet. See `docs/report/README.md` *for what it needs to cover.*